

Why AI Cannot Do Physics: a Counterclaim to the Generalizability of LLM Models to Physics Oriented Research

By James Squires

I've previously discussed why AI can do many parts of what is traditionally thought of as a mathematician's task quite effectively. However, I retained a distinction between the delegations of a physicist and a mathematician. The mathematician works with mathematical objects; in other words, they work in a closed, well defined system. This is inherently programmable. However, based on our current abilities, the same cannot be claimed for the role of a physicist.

When asked at a meeting to describe what physicists do, I said "Physicists observe physical phenomena and specify these into *physical* or *mathematical models*." In other words, we're translating senses and measurements observed in 'reality', whatever that may be, into a mathematical representation. We then analyze what this mathematical representation is capable of doing and where it breaks; this is called theoretical physics. The *philosopher stone* of physics is to obtain a mathematical representation that entirely matches physical reality. In other words, all predictions of the mathematical model are actual phenomena in our physical reality and all physical phenomena are predicted by the mathematical model. (notice that *physical model* and *mathematical model* are synonyms; these remain distinct from what I will call *physical reality*).

With this in mind, it seems quite obvious that the LLM is incapable of performing the role of a physicist: modeling physical reality (I will specifically discuss LLMs for this article). Unlike the mathematician, who works within a system, the physicist must specify one system to another. Specifically, the physicist specifies the system experienced by senses and measurement to one

entirely captured by mathematical representation. An LLM can easily work within a mathematical representation using the paradigm I laid out in our article on AI Math Proofs; however, an LLM has limited physics reasoning capacity.

The best simplification of an LLM is to think of it as a statistical prediction machine; it probabilistically returns the most expected results in some linguistic representation given some linguistic input. Now, what is a human? We actually process language from auditory or occipital senses. But we also think through our bodies... abstraction, particularly linguistic abstraction, is only one layer of nuance. To understand this, we should look at Piaget's psychology of child development.

1. **Sensorimotor Stage:** the infant or baby learns how to physically interact with objects and coordinate its senses. Some examples are gripping things, biting objects, poking things, seeing things.
2. **Preoperational Stage:** the toddler begins to acquire the basic abstractions of their perceptions. We can associate 'tree' to an actual tree, we associate the abstraction of a dinosaur to a dinosaur appearing on the TV screen.
3. **Concrete Operational Stage:** the child begins to learn rules or laws, they begin to be able to manipulate symbols; this produces more abstract reasoning and induction skills, although there are still some limitations.
4. **Formal Operational Stage:** the child/teenager can generate hypotheses, fully implement abstract reasoning. This is the stage where abstractions can stack and become more and more abstract and divorced from physical intuition.

Another way of viewing these stages is: basic building blocks → Basic abstraction → Abstract manipulation → Systemitization of abstraction. Notice that the fundamental foundation for abstraction is sensory perception. We can trace each abstraction or ‘token’ back to some physical perception, and with the abstractions, we can inductively construct seemingly endless more abstractions. LLMs do not have physical perception, nor does current technology enable LLMs to properly interface with physical reality; thus LLMs have abstraction without the anchoring of physical perception. Physical modeling, on the other hand, takes physical perception (measurement) to then relate and extend to a physical model. Modeling is the (iterative) process that carries physical perception to abstraction. This, at the present moment, is uniquely human. And I’d argue that, to invent an AI model capable of performing physical measurement, we essentially need to reinvent the human itself. Although we can partially semantically ground AI in physical reality by building a pipeline of physical perception (such as one AI uses convolutional neural networks to process images), though physical perception and the interplay between physical perception and abstraction is likely far more challenging to implement than the abstraction bit.